

Homework 13: Orthogonal Projection onto a Plane

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BCS A, E, F, G

Theorem 1. Construction of Orthogonal Projections. Suppose $F = \mathbb{R}$ or $F = \mathbb{C}$ and let S be an m –dimensional subspace of F^n . Then there is a unique orthogonal projection P onto S , and it is given by

$$P = AA^*,$$

where $A \in M_{n,m}(F)$ is a matrix with any orthonormal basis of S as its columns. ■

In the special case when S is 1–dimensional (i.e., a line), the above result simply says that $P = \mathbf{u}\mathbf{u}^*$, where $\mathbf{u}$ is a unit vector pointing in the direction of that line. It follows that $P\mathbf{v} = (\mathbf{u}\mathbf{u}^*)\mathbf{v} = (\mathbf{u} \cdot \mathbf{v})\mathbf{u}$, which recovers the fact that we noted earlier about functions of this form (well, functions of the form $P(\mathbf{v}) = \langle \mathbf{u}, \mathbf{v} \rangle \mathbf{u}$) projecting down onto the line in the direction of $\mathbf{u}$.

Elaboration. Assume that $F = \mathbb{R}$. We will discuss the case when $n = m = 2$. Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be an orthonormal basis such that

$$\mathbf{v}_1 = \begin{bmatrix} u_{11} \\ u_{21} \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} u_{12} \\ u_{22} \end{bmatrix}.$$

Then

$$A = \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}.$$

Assume that $F = \mathbb{R}$. Then

$$A^* = A^T = \begin{bmatrix} u_{11} & u_{21} \\ u_{12} & u_{22} \end{bmatrix}.$$

Now

$$AA^* = AA^T = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} \begin{bmatrix} u_{11} & u_{21} \\ u_{12} & u_{22} \end{bmatrix} = \begin{bmatrix} u_{11}u_{11} + u_{12}u_{12} & u_{11}u_{21} + u_{12}u_{22} \\ u_{21}u_{11} + u_{22}u_{12} & u_{21}u_{21} + u_{22}u_{22} \end{bmatrix}.$$

$$AA^T = \begin{bmatrix} u_{11}u_{11} & u_{11}u_{21} \\ u_{21}u_{11} & u_{21}u_{21} \end{bmatrix} + \begin{bmatrix} u_{12}u_{12} & u_{12}u_{22} \\ u_{22}u_{12} & u_{22}u_{22} \end{bmatrix} = \begin{bmatrix} u_{11} \\ u_{21} \end{bmatrix} [u_{11} \quad u_{21}] + \begin{bmatrix} u_{12} \\ u_{22} \end{bmatrix} [u_{12} \quad u_{22}] = \mathbf{v}_1\mathbf{v}_1^T + \mathbf{v}_2\mathbf{v}_2^T.$$

This shows that

$$AA^* = \mathbf{v}_1\mathbf{v}_1^* + \mathbf{v}_2\mathbf{v}_2^*.$$

One of the most useful features of orthogonal projections is that they do not just project a vector $\mathbf{v}$ anywhere in their range, but rather they *always project down to the closest vector in their range*.

Theorem 2. Another Construction of Orthogonal Projections.

Suppose V is an inner product space and $S \subseteq V$ is an n –dimensional subspace with orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$. Then the linear transformation $P: V \rightarrow V$ defined by

$$P(\mathbf{v}) = \langle \mathbf{u}_1, \mathbf{v} \rangle \mathbf{u}_1 + \langle \mathbf{u}_2, \mathbf{v} \rangle \mathbf{u}_2 + \dots + \langle \mathbf{u}_n, \mathbf{v} \rangle \mathbf{u}_n, \quad \forall \mathbf{v} \in V$$

is an orthogonal projection onto S .

Problem. Construct the orthogonal projection P onto the plane $S \subset \mathbb{R}^3$ with equation $x - y - 2z = 0$.

Hint. Use orthonormal basis of S from homework 12.

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Good Luck